

①

المطلوب الأول

① مثال (7) صفحة (5)

Ex/7//

$$x + \frac{1}{x} > 2$$

$$x + \frac{1}{x} - 2 > 2 - 2$$

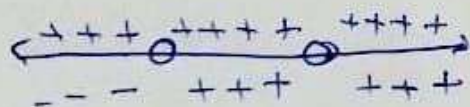
$$\underline{x} + \frac{1}{x} - 2 > 0$$

$$\frac{x^2 + 1}{x} - 2 > 0$$

$$\frac{x^2 + 1 - 2x}{x} > 0$$

$$\frac{x^2 - 2x + 1}{x} > 0$$

$$\frac{(x-1)^2}{x}$$



$$S = (0, 1) \cup (1, \infty)$$

$$\frac{(x-1)^2}{x} = 0$$

يجد الطرفين

$$x - 1 = 0$$

$$x = 1$$

$$x = 0$$

(2)

(المجموعة الأولى)

Ex/6

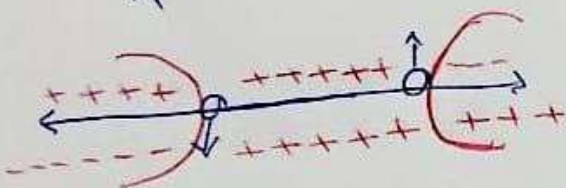
مثال 4 صفحة 4

$$\frac{1}{x} < 4$$

$$\frac{1}{x} - 4 < 4 - 4$$

$$\frac{1}{x} - 4 < 0$$

$$\frac{1-4x}{x}$$



$$S = (-\infty, 0) \cup (\frac{1}{4}, \infty)$$

$$1-4x=0$$

$$-4x=-1$$

$$4x=1$$

$$x=\frac{1}{4}$$

$$x=0$$

③

Ex/6/

$$21 + 4x - x^2 > 0$$

المطلوب الثانية مثال ⑥ صيغة ③

$$21 + 4x - x^2 > 0 \quad [-1]$$

$$x^2 - 4x - 21 < 0$$

$$(x - 7)(x + 3)$$

$$x = 7$$

$$x = -3$$



$$S = (-3, 7)$$

مثال ⑤ صيغة ②

Ex/5/

$$x^2 \geq 1$$

$$x^2 - 1 \geq 1 - 1$$

$$x^2 - 1 \geq 0$$

$$(x - 1)(x + 1) \geq 0$$

$$x = 1$$

$$x = -1$$



$$S = (-\infty, -1] \cup [1, \infty)$$

(4)

Ex/3/

ملزمة الثانية مثال (3) صفة (7)

$$x^2 - |x| = 12$$

[1]  $x \geq 0$

$$x^2 - x = 12$$

$$x^2 - x - 12 = 0$$

$$(x-4)(x+3)$$

$\therefore \boxed{x=4}$  or  $x=-3$  ~~يحل~~

$x < 0$

[2]  $\rightarrow x^2 - (-x) = 12$

$$x^2 + x - 12 = 0$$

$$(x+4)(x-3)$$

$x = -4$  or  $x = 3$  ~~يحل~~

$$S = (-4, 4)$$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$x = 0$$



(5)

$$EX/4 / \frac{4-X^2}{|2-X|} = 9$$

حلزوة الثانية مثال (4) صيغة (8)

$$|2-X| = \begin{cases} (2+X), X \geq 2 \\ -(2-X), X < 2 \end{cases}$$

1  $X \geq 2$

$$\frac{4-X^2}{2-X} = 9$$

$$\frac{(2-X)(2+X)}{(2-X)} = 9$$

$$\therefore 2+X = 9$$

$$\boxed{X = 7}$$

$$2-X = 0$$

$$\boxed{X = 2}$$

2  $X < 2$

$$\frac{4-X^2}{-(2-X)} = 9$$

$$\frac{(2-X)(2+X)}{-(2-X)} = 9$$

$$-2-X = 9$$

$$-X = 11 \Rightarrow \boxed{X = -11}$$

$$\frac{4-(-11)^2}{2-(-11)} = \frac{4-121}{13} = \frac{-117}{13} = -9$$

بالقويض

$$S = \{7\}.$$

(4)

$$-1 \leq |x^2 - 8| \leq 1$$

$$8 - 1 \leq x^2 - 8 + 8 \leq 1 + 8$$

$$7 \leq x^2 \leq 9$$

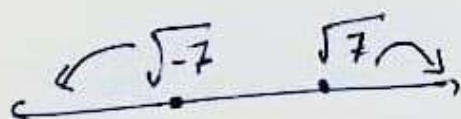
$$7 \leq x^2 \text{ and } x^2 \leq 9$$

$$x^2 \geq 7 \text{ and } x^2 \leq 9$$

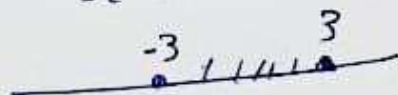
$$|x| \geq \sqrt{7} \wedge |x| \leq 3$$

$$x \geq \sqrt{7} \text{ or } x \leq -\sqrt{7}$$

$$S_1 = (-\infty, \sqrt{7}] \cup [\sqrt{7}, \infty) \wedge -3 \leq x \leq 3$$



$$S_2 = [-3, 3]$$



$$S = (-\infty, -\sqrt{7}] \cup [\sqrt{7}, \infty) \cap [-3, 3]$$



$$S = [-3, -\sqrt{7}] \cup [\sqrt{7}, 3]$$